

EES203: Geochemistry

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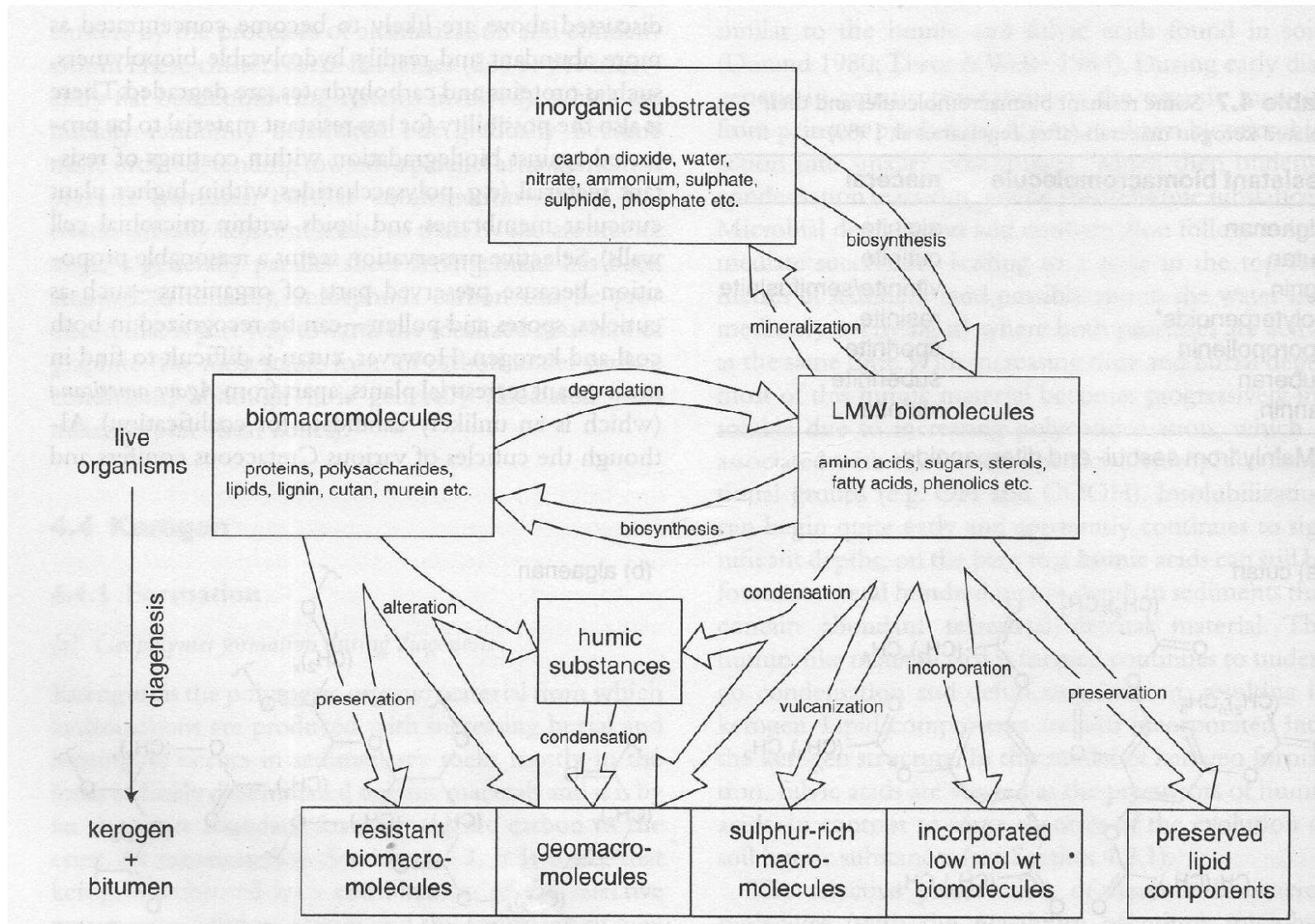
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Organic geochemistry

Kerogen

- Polymeric organic material from which hydrocarbons are produced with increasing burial and heating
- Most abundant form of organic carbon in the earth crust
- Formed by a combination of the selective preservation of polymers and the formation of new geopolymers

Formation of Kerogen



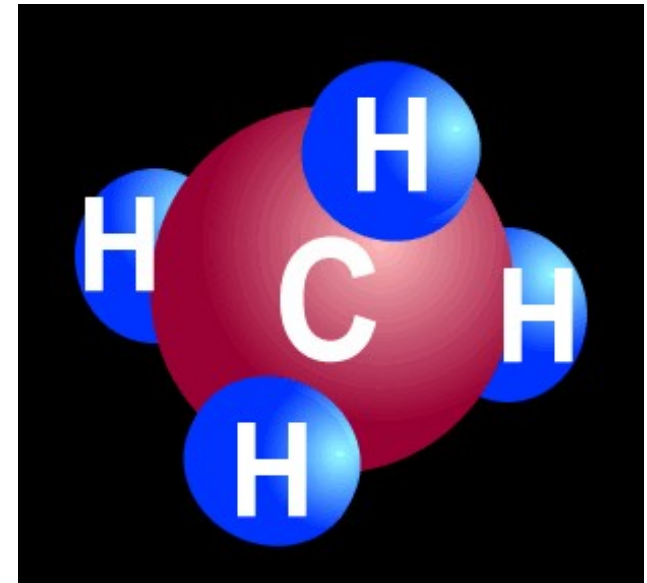
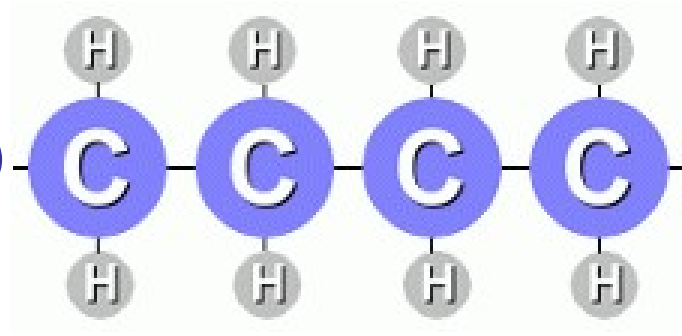
Simplified model for kerogen formation incorporating classical and selective preservation models

Petroleum System

- **Source (kerogens)**
- **Generation (Maturation)**
- **Migration**
- **Trap**
- **Reservoir**

What is petroleum?

- **Petroleum:** A general term for all naturally occurring hydrocarbons (hydrogen + carbon)
- **Solid Hydrocarbons:** Asphalt
- **Liquid Hydrocarbons:** Crude oil
- **Gas Hydrocarbons:** Natural Gas: methane, butane, propane, etc.



The simplest hydrocarbon is Methane (CH₄)

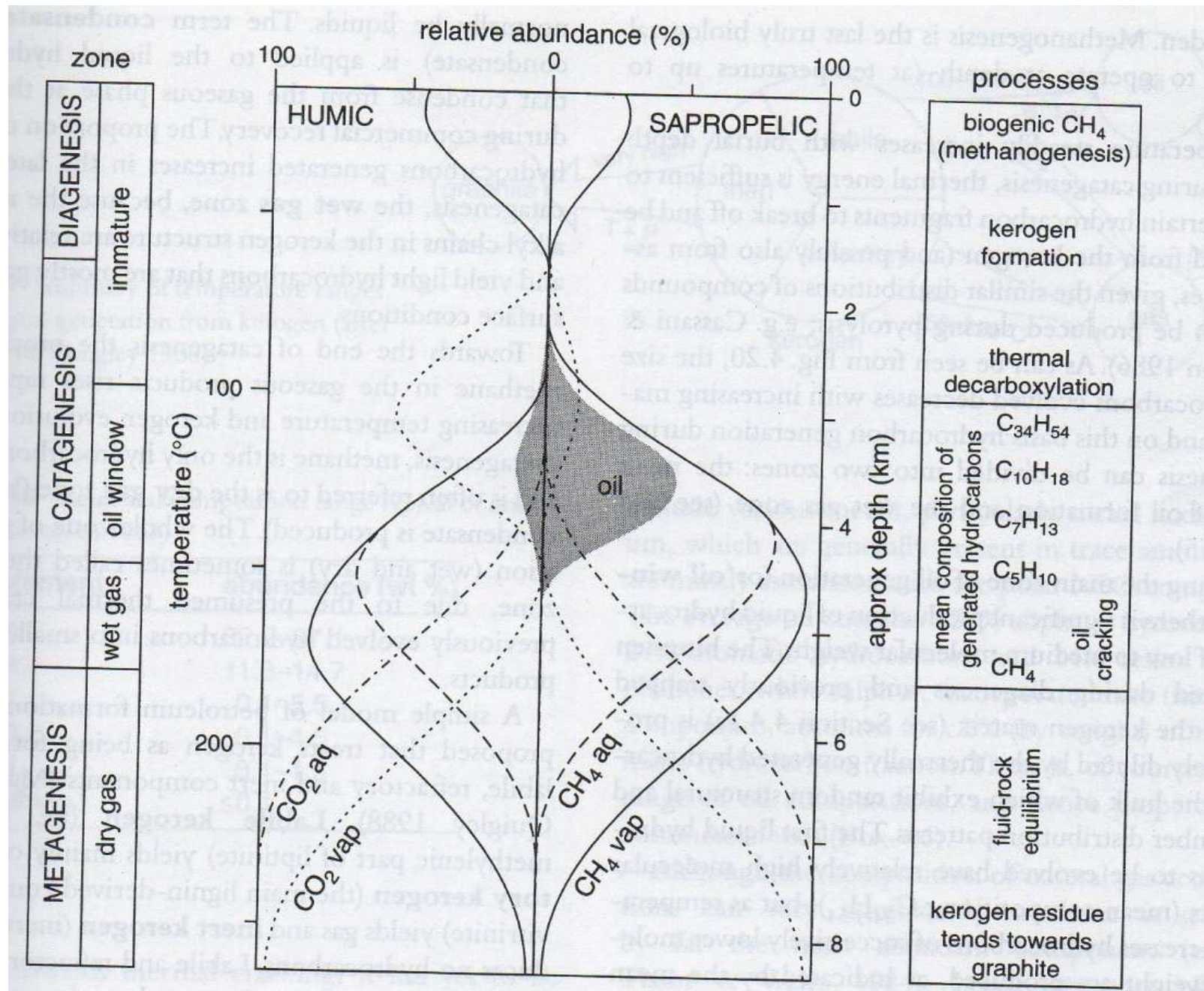
Types of hydrocarbons

- Includes oil and gas in various reservoirs, bitumen in oil sands and clathrate hydrates.
- The major gaseous hydrocarbons are the alkanes methane (natural gas, CH_4), ethane (C_2H_6), propane (C_3H_8), and butane (C_4H_{10}).
- Liquid components of petroleum include a complex mixture primarily of linear and cyclic hydrocarbons from C_5 to C_{17} , as well as numerous other molecular species
- Solid hydrocarbons include such broad categories as paraffin waxes (typically from C_{18} to C_{40}).

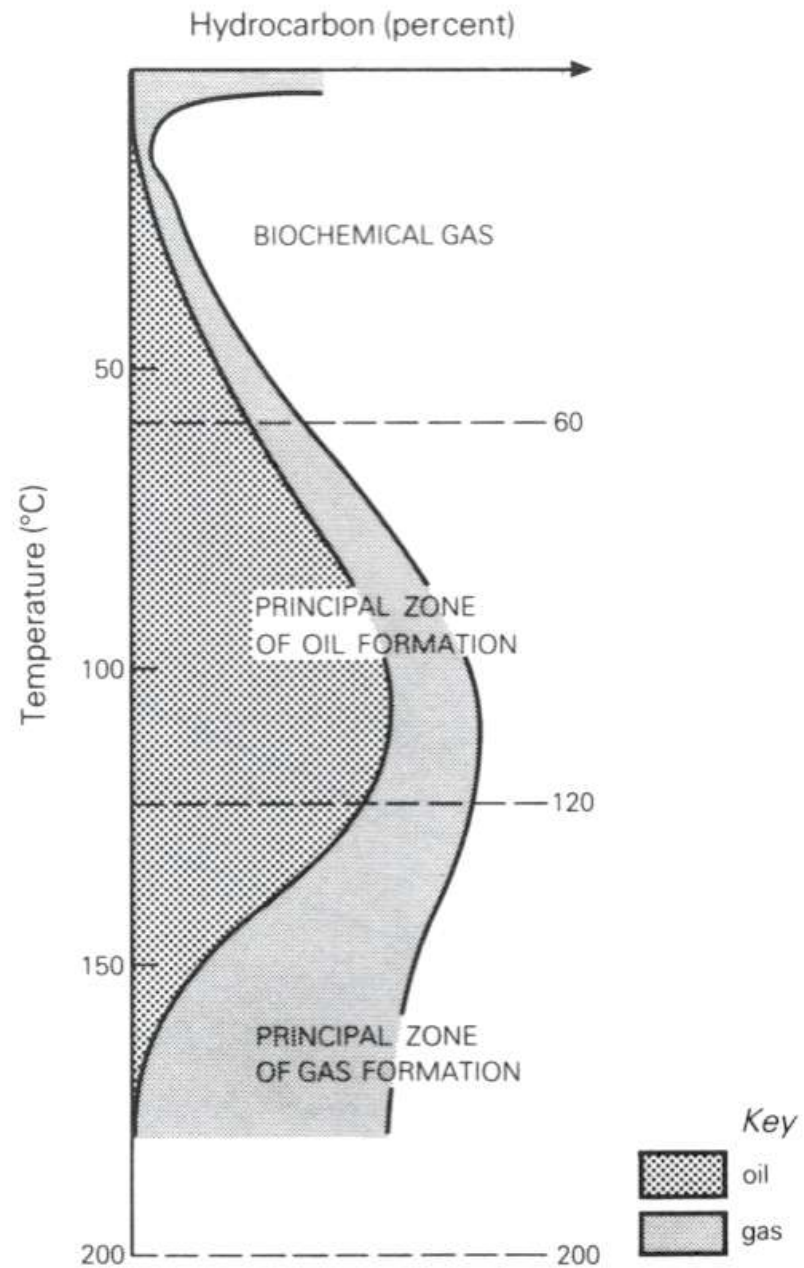
Origin of petroleum

- Hydrocarbon rich fluids evolved from Kerogen during catagenesis and metagenesis
- The amount and composition of the petroleum generated depend on the amount and composition of organic matter incorporated into the kerogen.

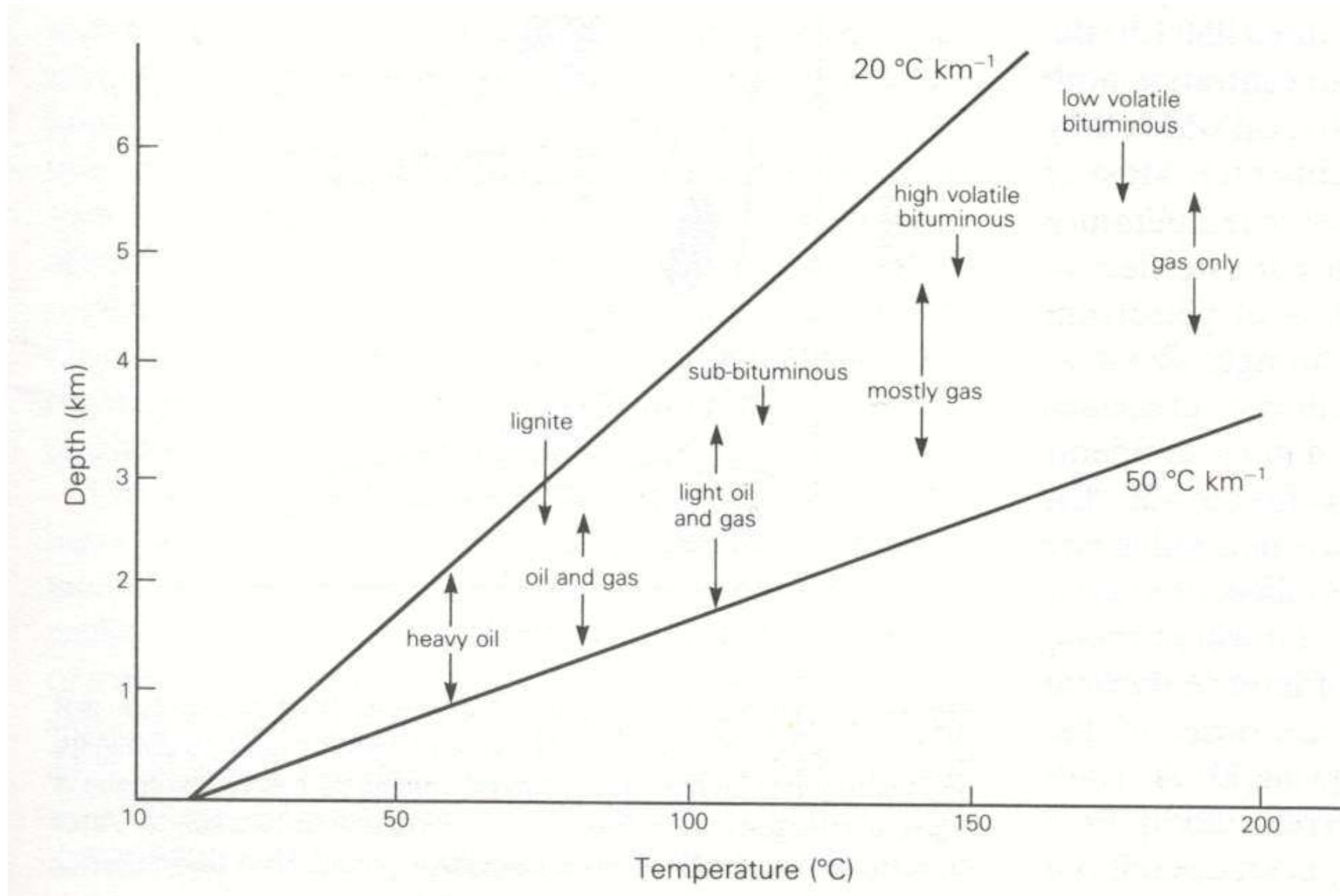
Petroleum generation



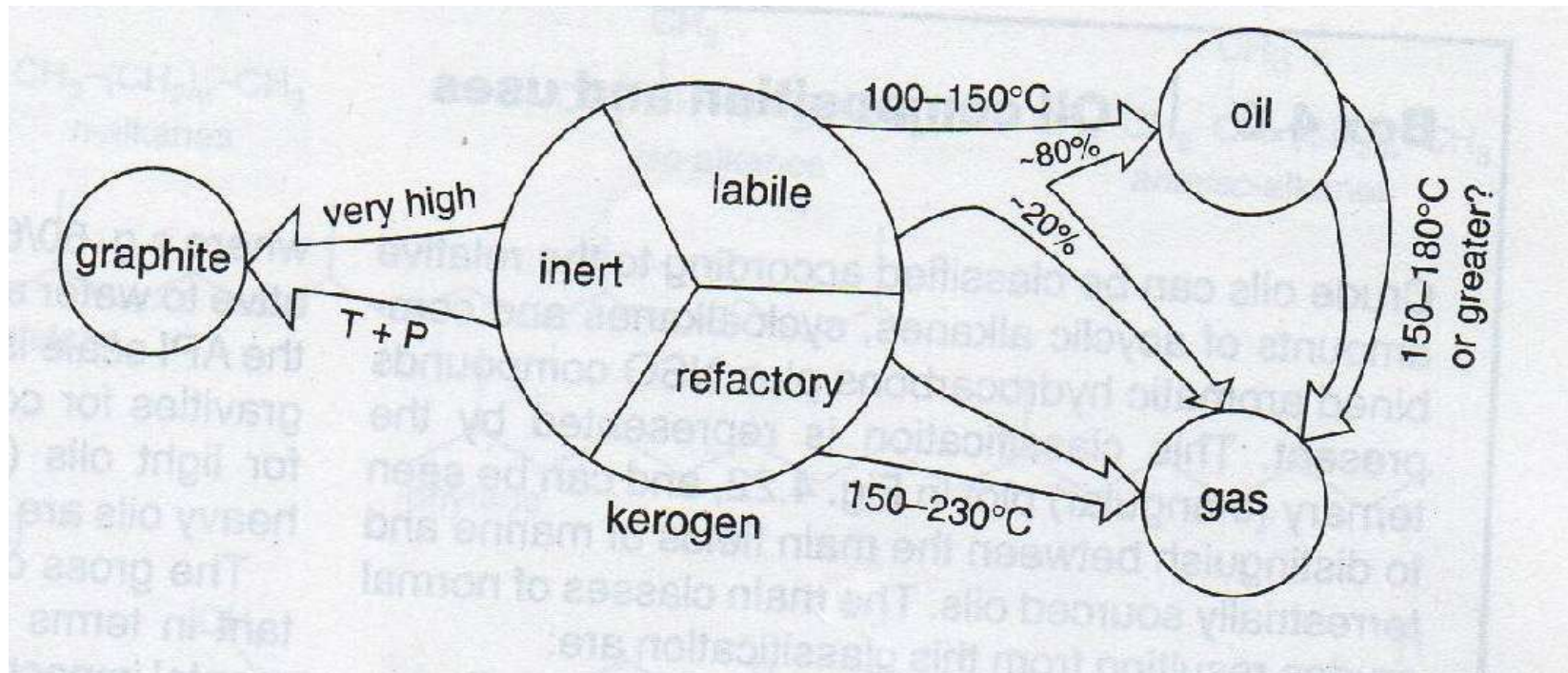
Petroleum generation



Summary of temperature ranges for Petroleum generation

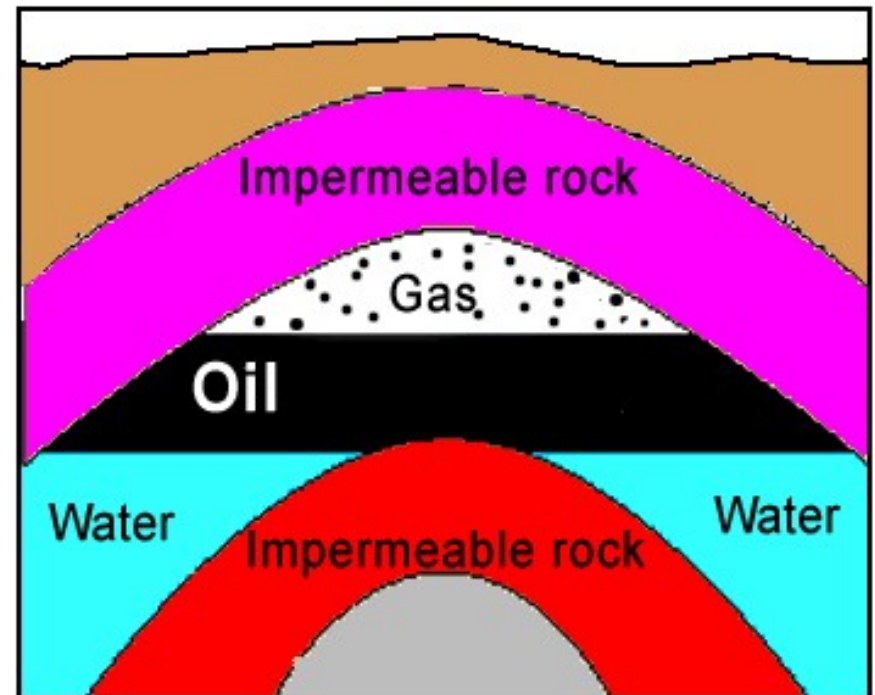


Summary of temperature ranges for Petroleum generation



Traps

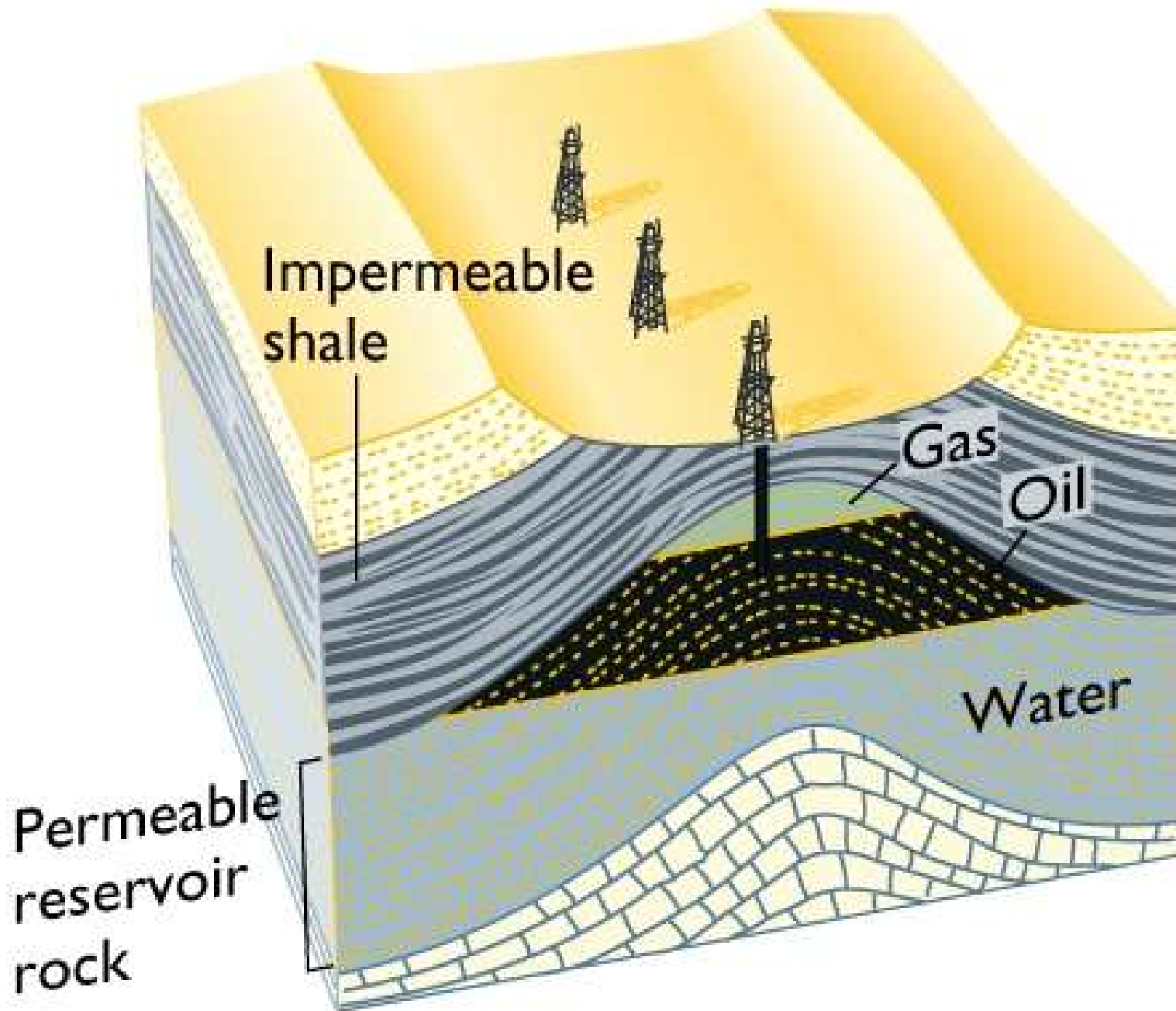
- If nothing stops oil from rising, it will reach surface
 - Ex: The La Brea tar pits
- Traps can be rocks that do not allow fluids to pass through them, or folds and faults in the rock can trap petroleum



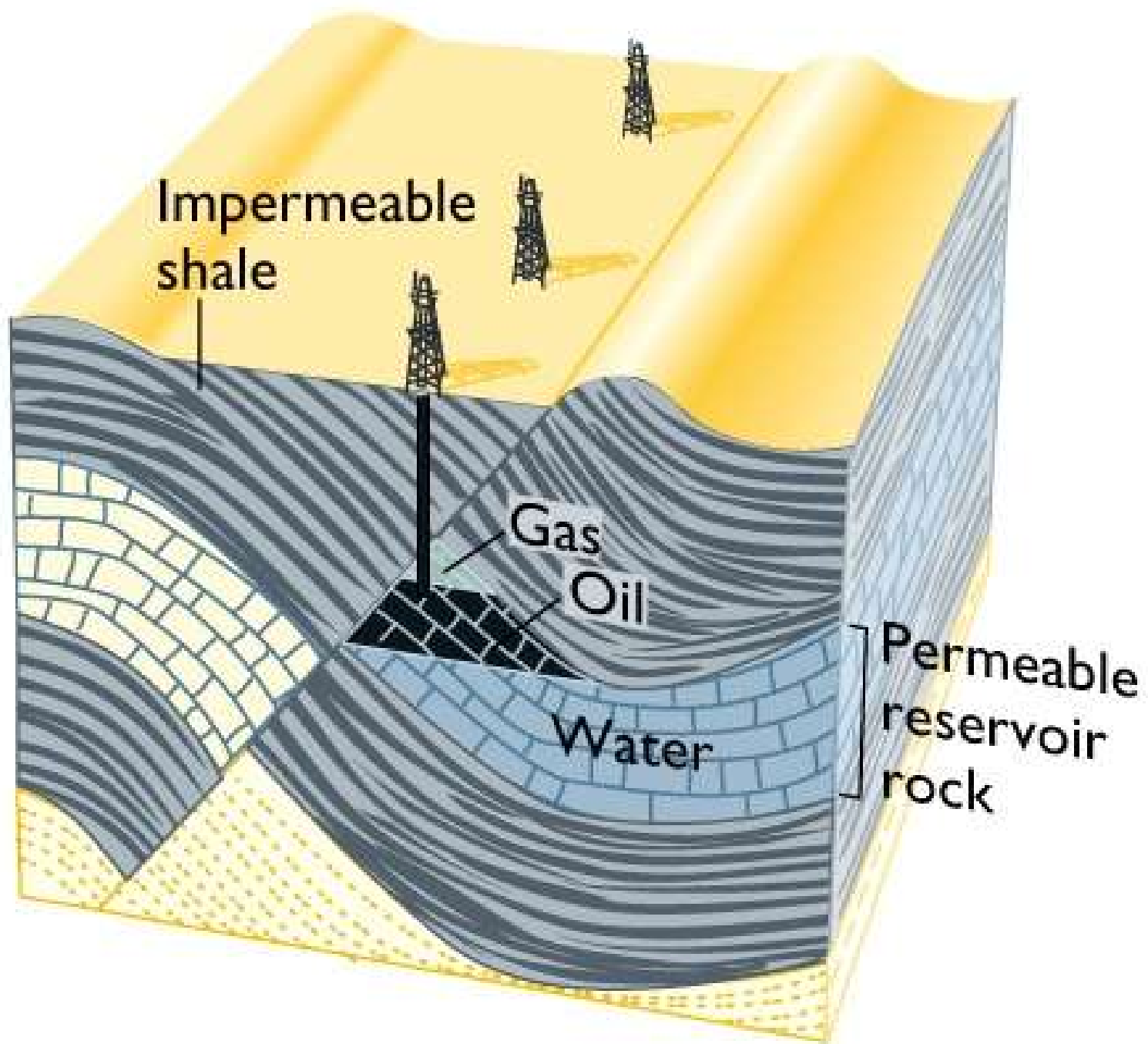
Trap

- Three dimensional geological feature that obstructs the flow of petroleum, forming a accumulation
- Two types of traps (i) Structural
 - (ii) Stratigraphical

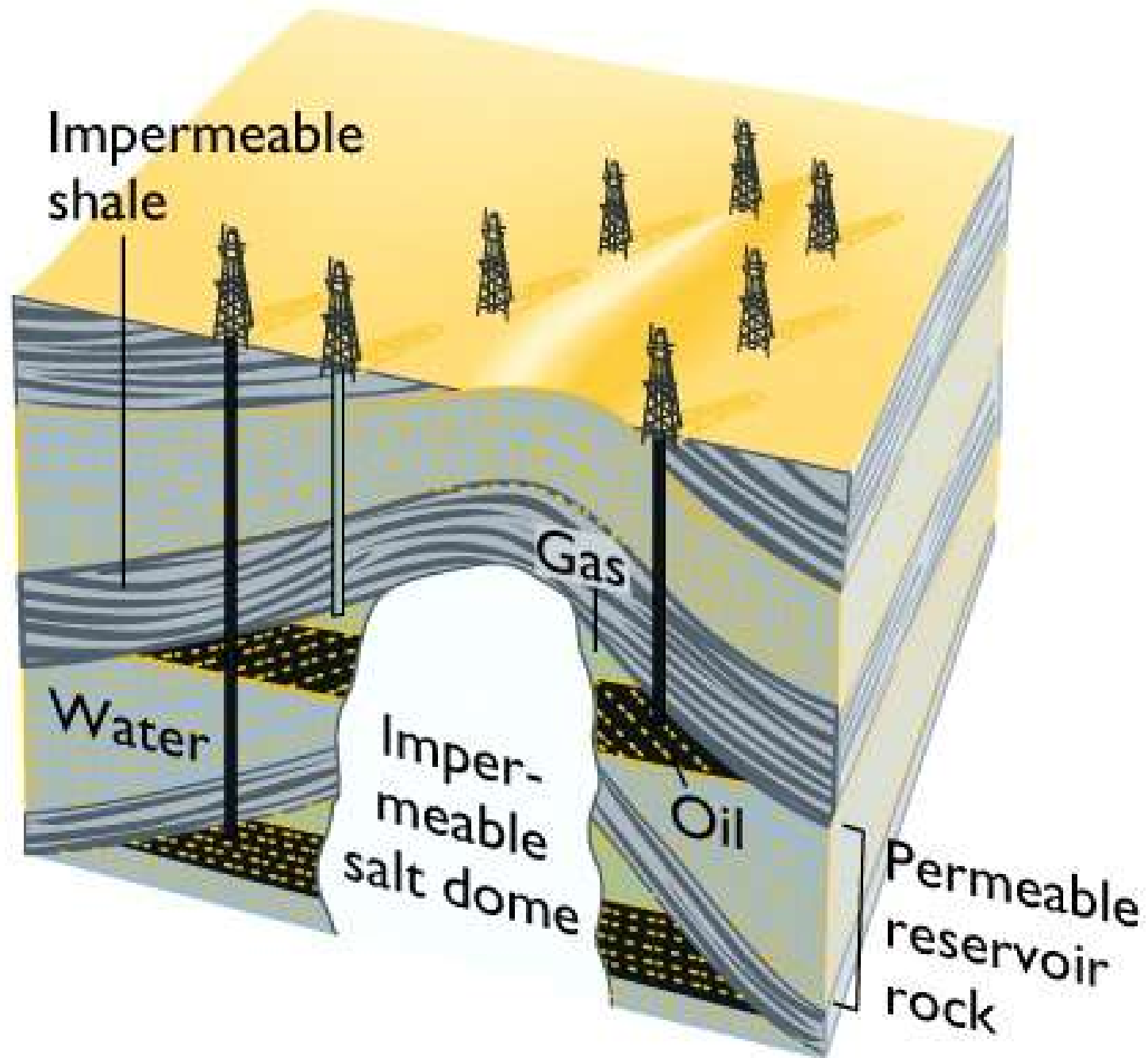
Structural trap: Anticline



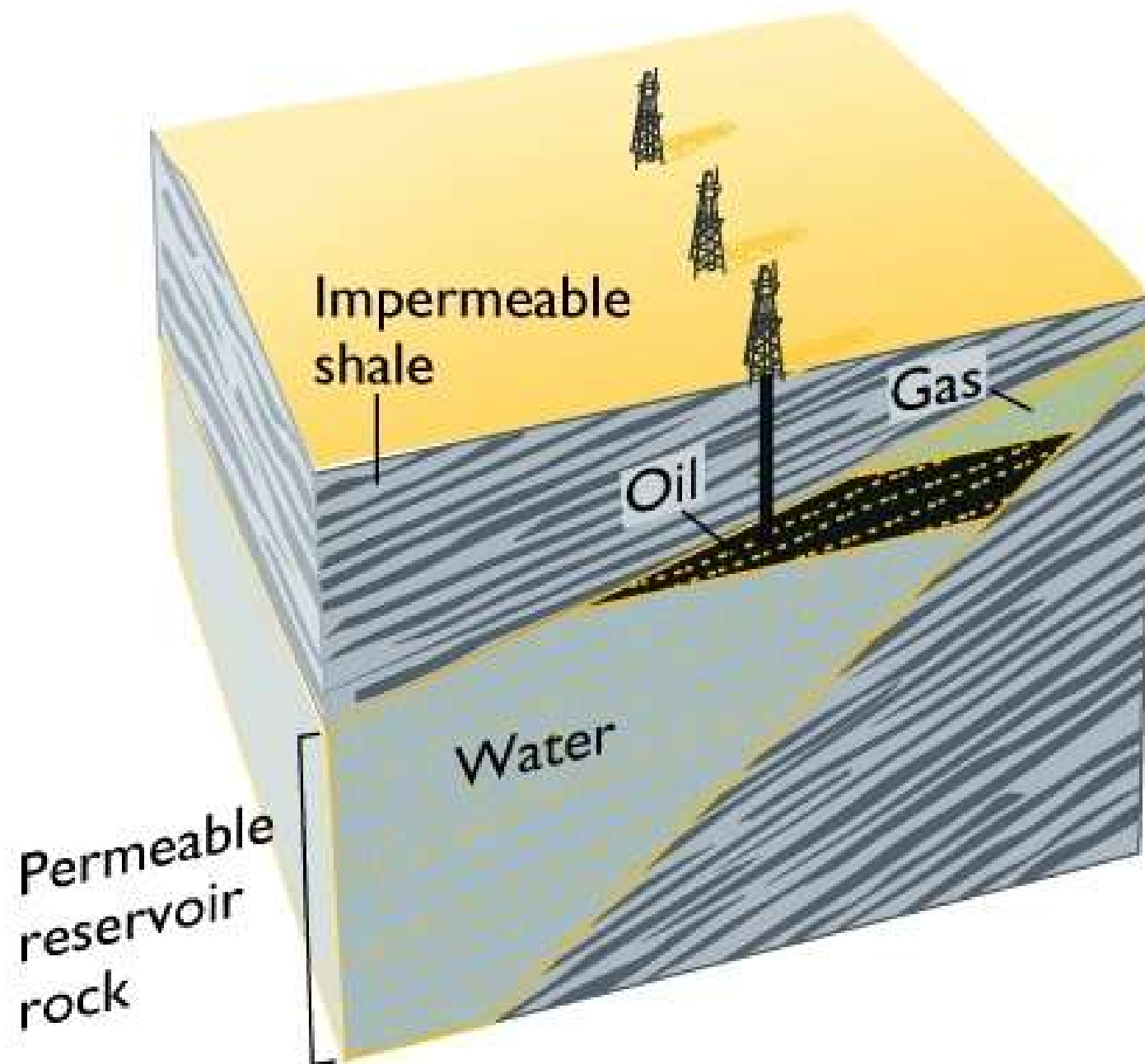
Structural trap: Fault



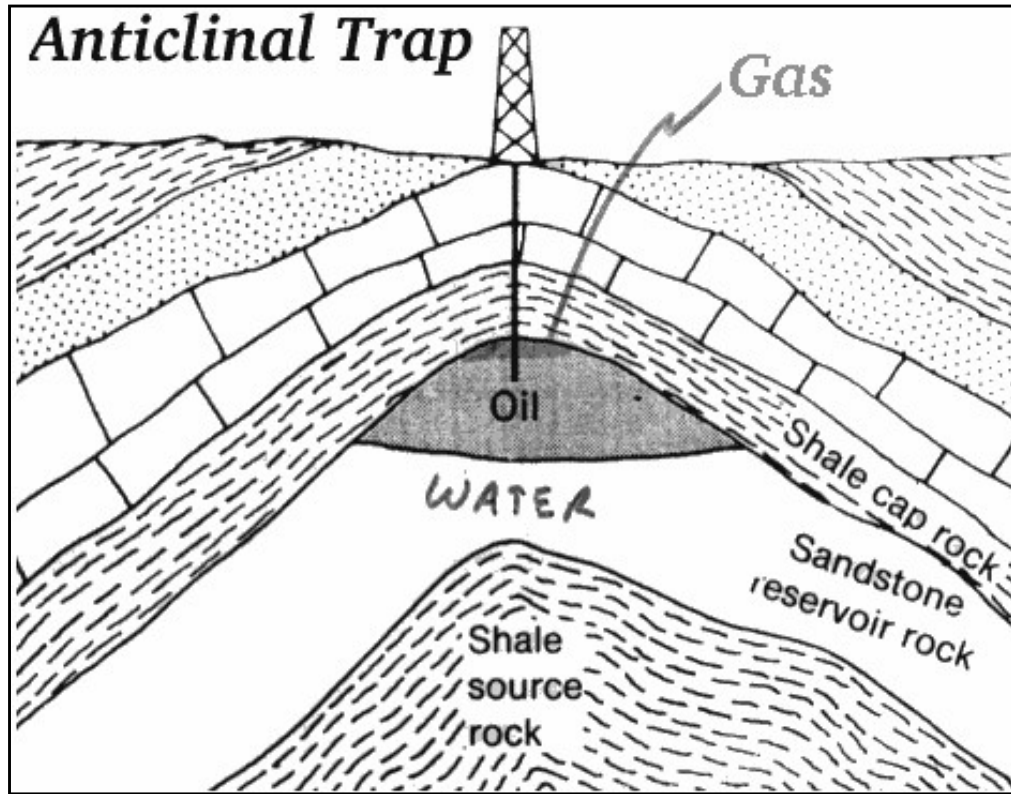
Structural trap: Salt dome



Stratigraphic Trap



Reservoir rocks



- A good reservoir rock is:
 - Porous: holes
 - Permeable: holes are connected
 - so that its fluids can be produced (removed from them)

Reservoir rocks

The two principal properties required from a rock to be a viable reservoir rock are **porosity and permeability**.

- Porosity is the capability of a rock to hold fluids in pores. It is expressed as a volume percent of the total rock and can range from very low porosities (a few %) to very high (over 40% in some chalks). Pores can be of many types, particularly in carbonate rocks.

- Permeability is the capability of a rock to transmit a fluid. It depends crucially on the connections between the pores. Darcy's law establishes the basic relationship between pressure, flow rate and permeability.

Reservoir rocks

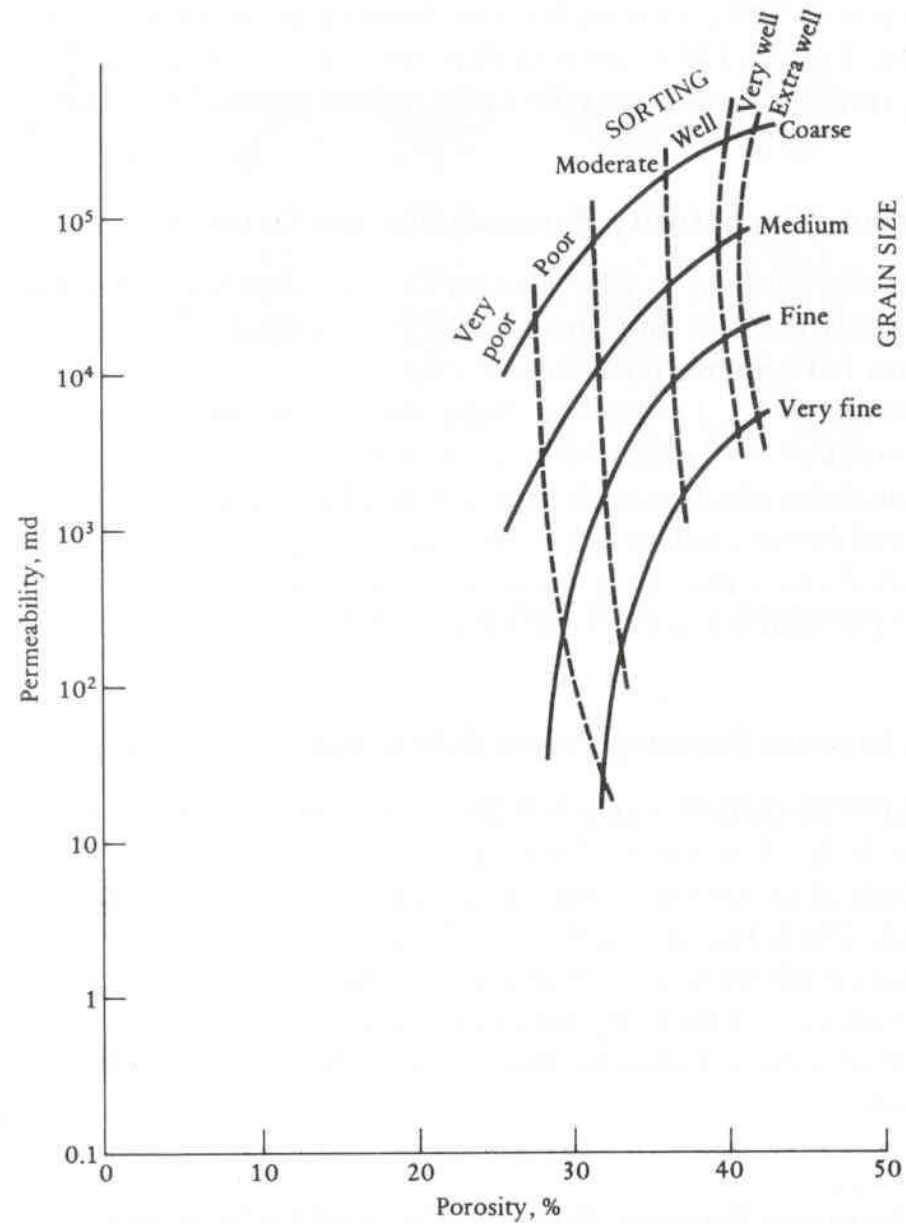
Major Factors Affecting k

In clastic rocks, the three-dimensional pore network is a function of the grain properties (the texture).

Grain size is probably the most important factor affecting permeability. Small grains generally have smaller pores and smaller pore throats than larger ones; fine-grained sandstones are therefore usually lower in permeability than coarse-grained ones.

Grain sorting is another important factor controlling permeability. If the grain distribution is very wide, the smaller pores can more easily block the pore throats and therefore reduce permeability.

Reservoir rocks



Maturation Indicators

- To recognize and evaluate potential source rocks for oil and gas by measuring their contents in organic carbon and their thermal maturities
- To correlate oil types with probable source beds through their geochemical characteristics and the optical properties of kerogen in the source beds
- To determine the time of hydrocarbon generation, migration and accumulation
- To estimate the volumes of hydrocarbons generated and thus to assess possible reserves and losses of hydrocarbons in the system.

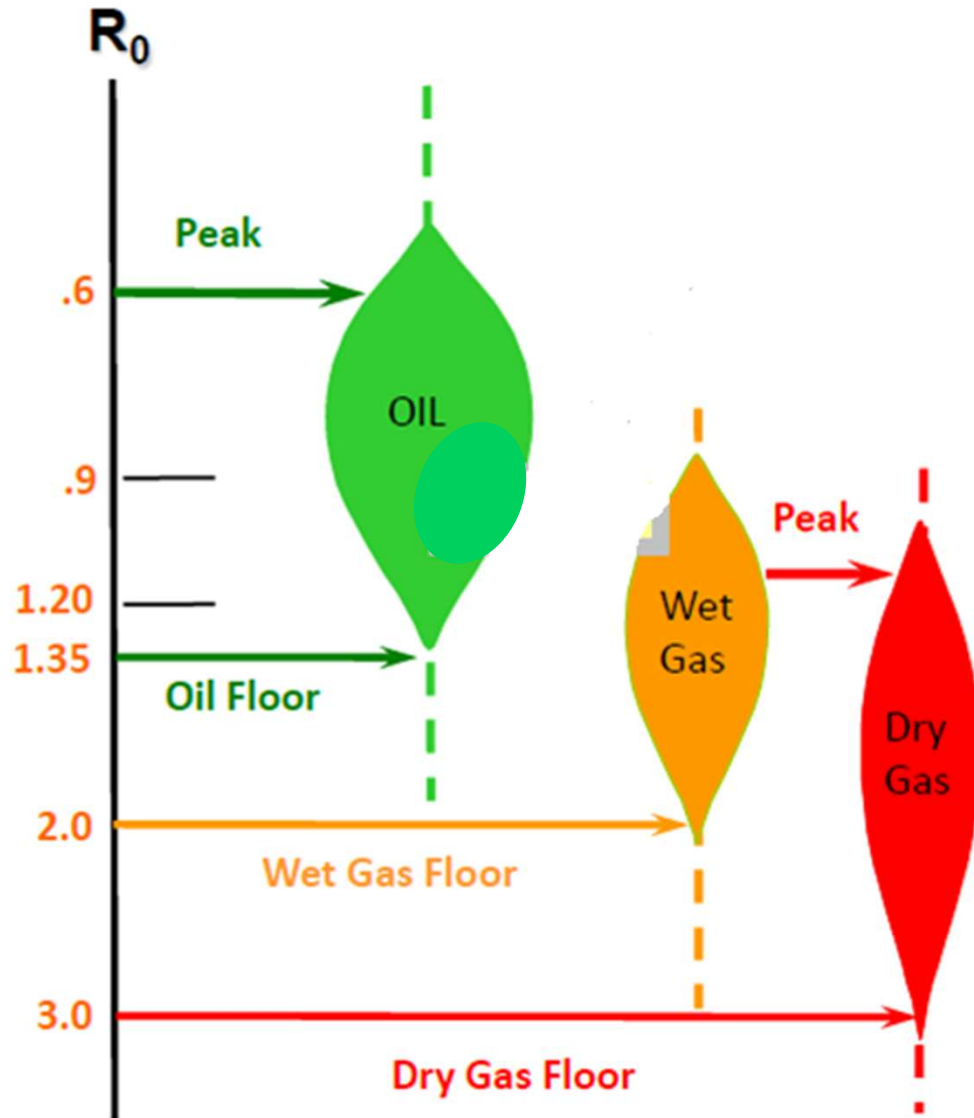
Total Organic Carbon (TOC) measurements

If a rock contains significant amounts of organic carbon, it is a possible source rock for petroleum or gas. The TOC content is a measure of the source rock potential.

The table below shows how TOC (in weight percent) relates to the source rock quality.

TOC Quality	
0.0-0.5	poor
0.5-1.0	fair
1.0-2.0	good
2.0-4.0	very good
>4.0	excellent

Vitrinite reflectance



R_0 : measures intensity of reflected light from polished vitrinite particles and reported as mean percent reflectance - the most commonly used technique for source rock thermal maturity determination.

Predictor of petroleum potential in source rock, a measure of thermal maturation

Calibrated to reflectance of crude oil;

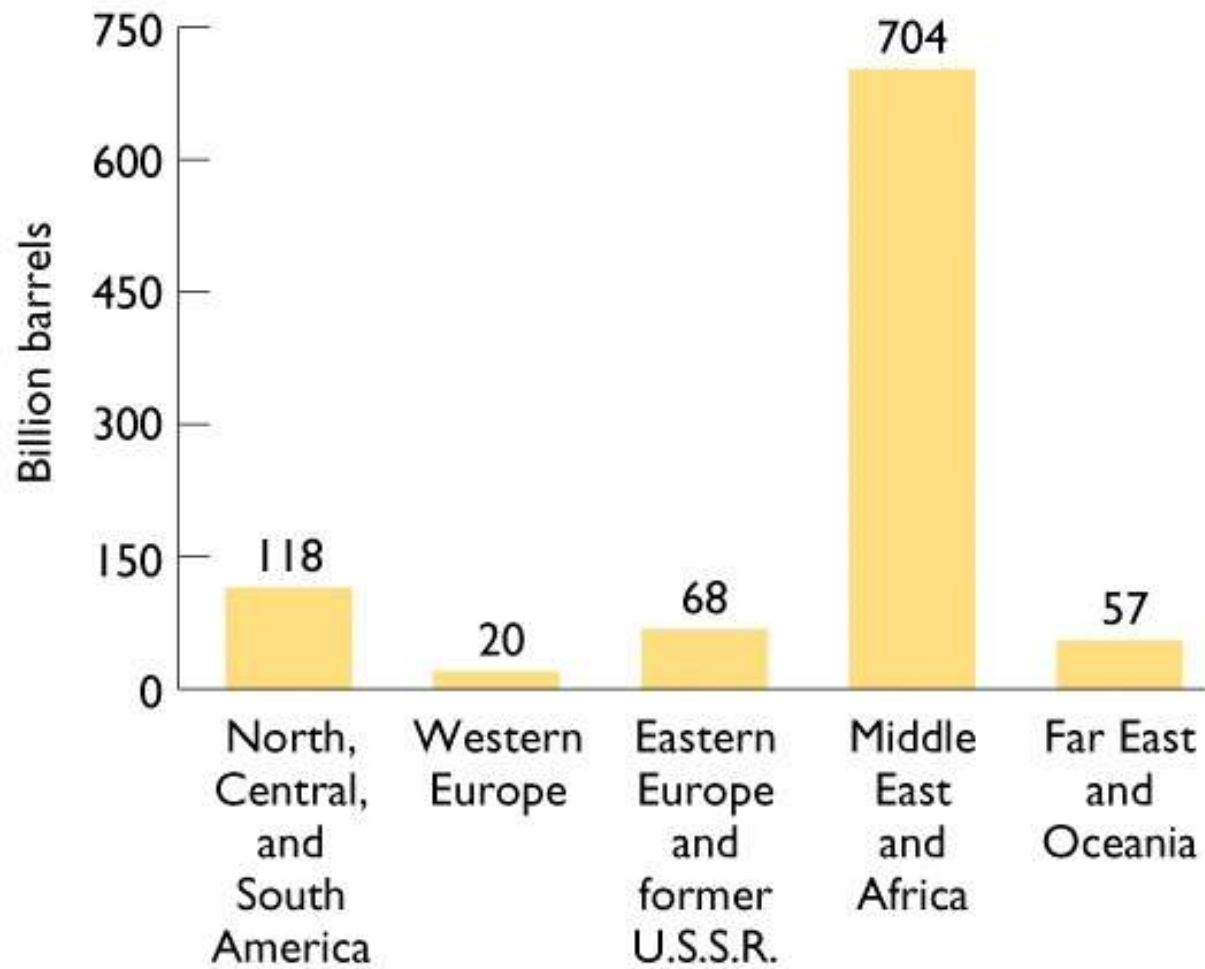
R_0 index = **1.0 (fully mature hydrocarbon)**

$R_0 > 0.8$ thermally mature organic matter, oil and gas present

$R_0 = 0.55-0.8$, organics in petroleum/gas producing range

$R_0 < 0.55$ thermally immature organic matter

Reserve of crude oil

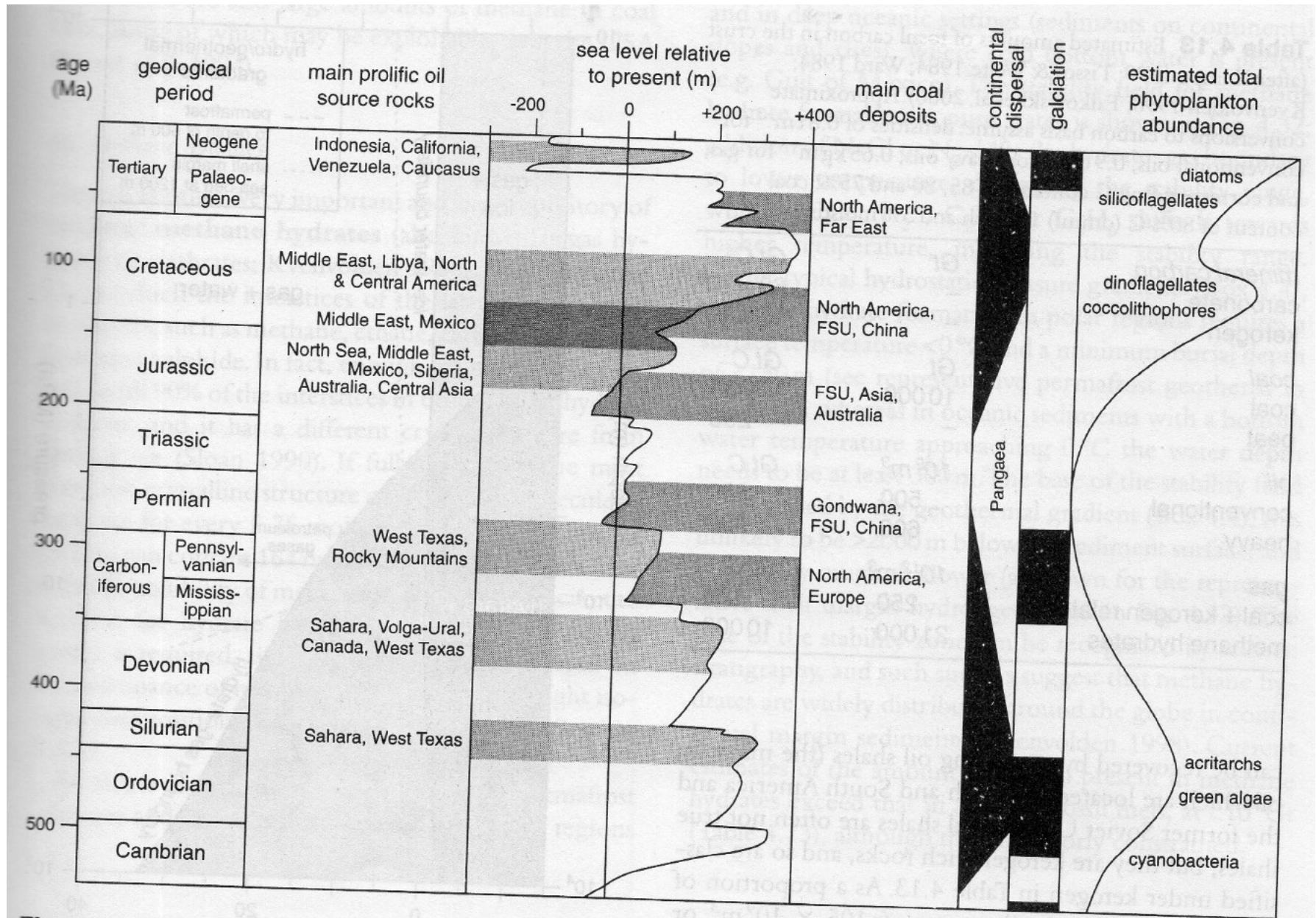


Temporal and Spatial distribution of Fossil organic carbon

Oil and gas

- **The major prolific oil source rocks were deposited during global marine transgressions.**
- **Two major episodes of oil formation can be distinguished**
 - : Devonian (Paleozoic)**
 - : Cretaceous**
- **The majority of oil producing peak coincides with the high phytoplankton abundance**

Eustatic sea level and major depositional period of coal and petroleum



Abiotic origins of deep hydrocarbons

Abiotic origins of deep hydrocarbons

Deep gas theories

- The hypothesis that at least some components of petroleum have a **deep abiotic origin in the lower crust or mantle**
- This abiotic origin has been proposed by Russian chemist and mineralogist **Dimitri Mendeleev** (Mendeleev 1877) and astronomer and mathematician **Fred Hoyle** (Hoyle 1955).

Abiotic origins of deep hydrocarbons

Deep gas theories

- The deep-Earth gas hypothesis proposes that abiogenic methane reflects a cosmic organic inheritance that is subsequently released by the mantle and migrates towards the surface utilizing weaknesses in the crust.
- The deep sourced methane polymerizes en route to higher molecular weight hydrocarbons that ultimately form petroleum deposits.

Abiotic origins of deep hydrocarbons

Deep Hot Biosphere hypothesis

- The abiotic petroleum hypothesis first gained significant exposure outside the Soviet Union during the 1977 gasoline crisis, when Cornell astrophysicist Thomas (Tommy) Gold published an editorial in the *Wall Street Journal* (Gold 1977).
- Most deep hydrocarbons are generated abiotically in the mantle and migrate to the crust, where they act as an energy source for microbes producing a deep microbial ecosystem—the “Deep Hot Biosphere”
- The organic remains of microbes in the deep hot biosphere represent the source of biological molecules in petroleum as a biological overprint onto an abiogenic organic mixture.

Evidences for abiotic origin of petroleum

- The presence of clearly abiotic hydrocarbons and other organic molecules in meteorites and on other solar system bodies
- Plausible synthetic pathways for mantle hydrocarbon production
- The association of hydrocarbon deposits with helium and other trace gases, presumably from mantle sources
- The existence of extensive deep microbial communities that impart an overprint of biomarkers onto the abiotic petroleum

Evidence for abiotic hydrocarbon synthesis

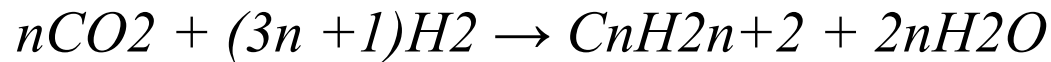
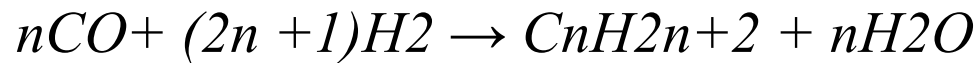
Abiotic hydrocarbons in space

- The greatest amounts of hydrocarbons in our solar system are associated with the gas giant planets and their satellites.
- The prevalence of non-biological hydrocarbons in the early stages of our solar system is evidenced by the chemistry of carbonaceous meteorites.
- Carbonaceous meteorites contain percentage levels of organic matter that were generated in the absence of biology.
- The widespread presence of organic matter in the Cosmos suggests that no exotic mechanism therefore need be invoked to propose abiogenic hydrocarbons on Earth.

Evidences for abiotic origin of petroleum

Experimental hydrocarbon synthesis

The Fischer-Tropsch synthesis (Fischer and Tropsch 1926) involves the conversion of carbon monoxide or carbon dioxide to hydrocarbons

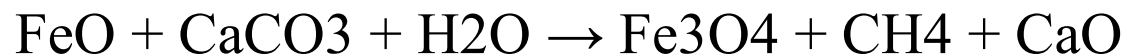


The Fischer-Tropsch process has been proposed as a generator of hydrocarbons in hydrothermal vent settings

Evidences for abiotic origin of petroleum

Experimental hydrocarbon synthesis

Simple **thermal metamorphism of carbonates** such as calcite (CaCO_3), dolomite [$\text{CaMg}(\text{CO}_3)_2$], and siderite (FeCO_3) at 400 °C in the presence of H_2 can produce low molecular weight hydrocarbons (Giardini and Salotti 1969), as in the following unbalanced reaction:



Evidences for abiotic origin of petroleum

Experimental hydrocarbon synthesis

- Hydration of olivine and can combine with carbon dioxide to form methane (Abrajano et al. 1990), as follows (unbalanced reaction):



Evidences for abiotic origin of petroleum

- The tendency of hydrocarbon reservoirs to occur at many depths in a single locality, implying an underlying deep source
- The distribution of hydrocarbon deposits related to underlying mantle structures
- The distribution of metals and other trace elements in petroleum, which correlate more strongly with chondrite compositions than presumed crustal sources

Evidence for abiotic hydrocarbon synthesis

Faults and plate margins

- The association of petroleum deposits with the putative conduits for upward migration, namely convergent plate margins and major fault zones (Gold and Soter 1980).
- Plate boundaries are associated with high geothermal gradients that enhance the formation, migration, and entrapment of oil and gas (Klemme 1975).

Evidence for abiotic hydrocarbon synthesis

Crystalline rock reservoirs

- The association of hydrocarbons with igneous rocks has been taken as prima facie evidence for abiogenic synthesis.
- A number reservoirs are situated in igneous rocks and, although most are relatively small, some giant fields (recoverable reserves > 500 million barrels of oil equivalent) do exist (Schutter 2003).

Evidence for abiotic hydrocarbon synthesis

Crystalline rock reservoirs

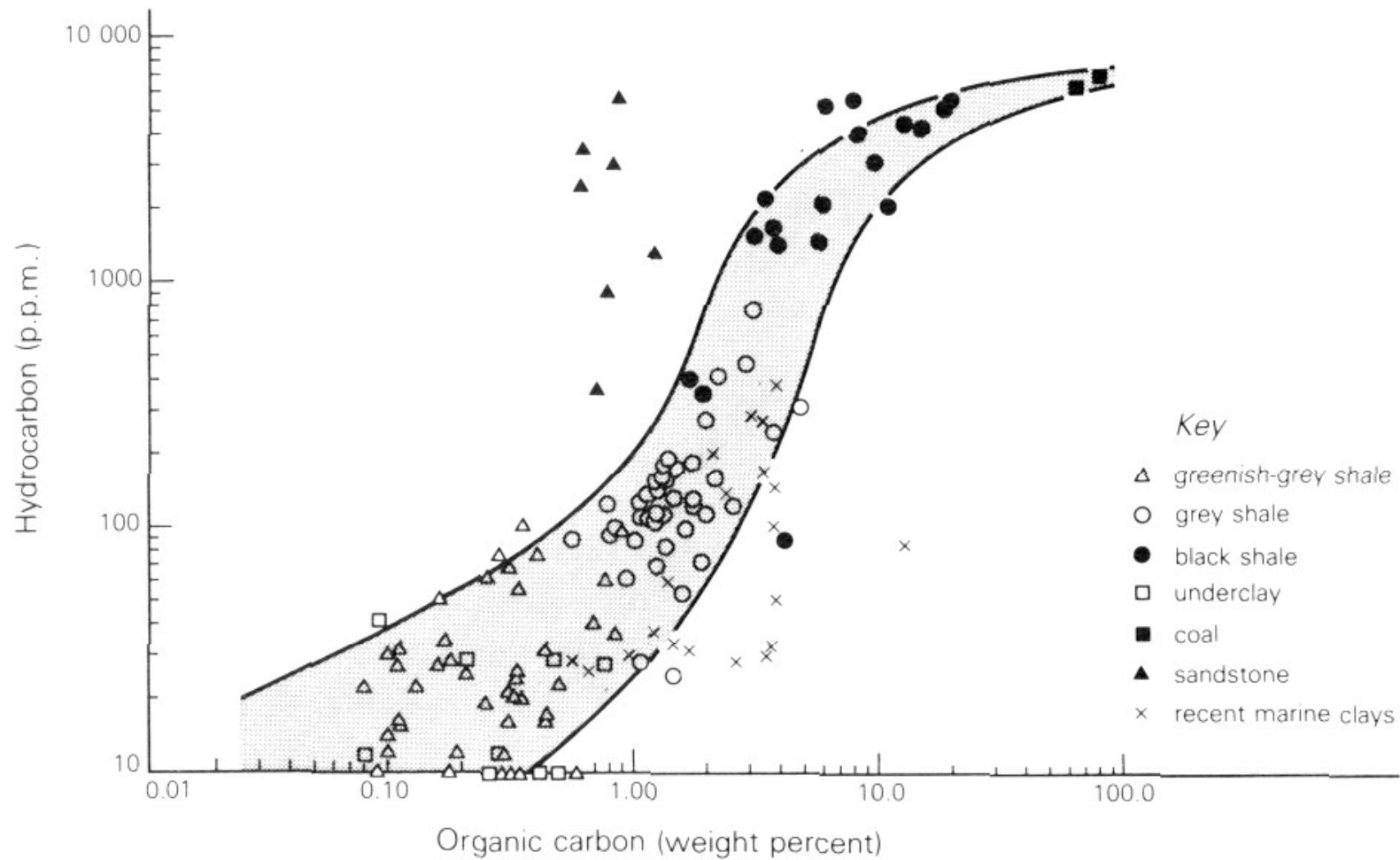
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Evidence for biotic hydrocarbon synthesis

Evidences for biotic origin of petroleum

- The crystalline rock reservoirs represent less than 1% of the world's petroleum deposits. The remaining 99% points to an overwhelming correlation of petroleum with sedimentary rocks, indicating a likely origin from sedimentary organic matter (Ulmishek and Klemme 1991).

Evidences for biogenic origin



- The relationship (hydrocarbon vs organic carbon) is a major argument for the biogenic origin of fossil hydrocarbons

Evidences for biotic origin of petroleum

Association with temperature and source rocks

- Petroleum deposits are associated with areas in which source rocks have been subjected to elevated temperatures in the subsurface.

Geochemical fingerprints; petroleum seeping from reservoirs reveals stable isotope ratios and molecular biomarkers that correlates it with thermally matured source rocks (Seifert and Moldowan 1978).

- The composition of hydrocarbons found in commercial oil fields have a low content of ^{13}C isotopes, similar to that found in marine and terrestrial plants; whereas hydrocarbons from abiotic origins such as methane have a higher content of ^{13}C isotopes.

Evidences for biogenic origin

Mantle oxidation state

- For abiogenic hydrocarbons to be produced from the mantle, conditions must be reducing to prevent the oxidation of hydrogen-rich organic compounds such as methane.
- Data suggest that Earth's upper mantle is too oxidized to allow the persistence of more than small amounts of methane or hydrogen (Manning et al. 2013).

Coal

- A sedimentary rock that burns
- Mineralized vegetative material deposited over a long period of time
- altered chemical composition
- Formed by increased T and P
- Partial decay resulting from restricted access to oxygen

Formation of Coal

Coal formation depends on two factors:

- (1) initiation, maintenance, and repetition of environments that favor large-scale accumulation and preservation of vegetal sediment
- (2) geochemical processes that induce chemical coalification of the peat to higher-rank coal.”

Source: DiCiccio, 1996

Classification of Coal

Humic coal: vascular plants remains

exhibit stratification and tend to be bright

Sapropelic coal: fine grained organic muds in quiet, oxygen deficient shallow waters

not stratified macroscopically and dull

Macerals of coal : morphological constituents

maceral group	maceral	origin
vitrinite	telinite	cellular structure of wood, leaf and root tissue
	collinite	structureless, infilling gel
	vitrodetrinite	unidentified cell fragments
liptinite	sporinite	spore and pollen cases
	cutinite	waxy coating of leaves and stems
	suberinite	cork tissues, e.g. bark and root walls
	resinite	resin bodies
	alginite	algal tests
	liptodetrinite	unidentified liptinite fragments
	fluorinite	lenses/layers, possibly plant essential oils
	bituminite	wisps or groundmass, from lipids
	exudatinitite	veins of expelled bitumen-like material
inertinite	fusinite	charred wood and leaf tissue
	semifusinite	partially charred wood and leaf tissue
	macrinite	charred gel material
	micrinite	charred liptinitic material
	sclerotinite	fungal remains (e.g. sclerotia)
	inertodetrinite	unidentified inertinite fragments

Hard (bituminous) coal macreal groups (modified Stopes-Heerlen system)

Formation of coal

Two main phases of formation of humic coals

- a) Peatification – biological
- b) Coalification- biochemical and geochemical stage

Rank of coal

How far a coal has progressed through peatification/coalification

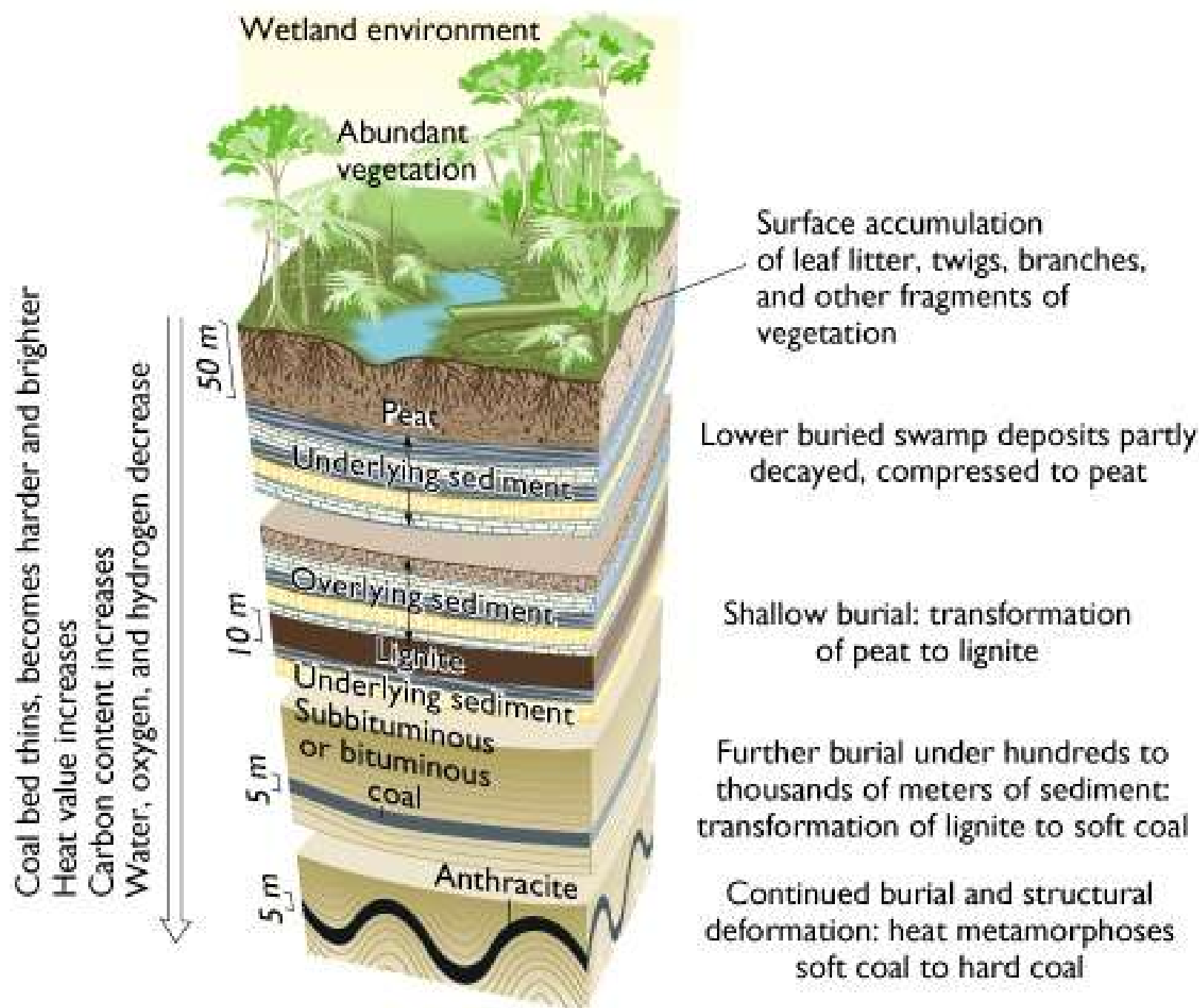
The main stages of humic coal formation are

1. Peat
2. Lignite
3. Sub-bituminous coal
4. Bituminous coal
5. Anthracite

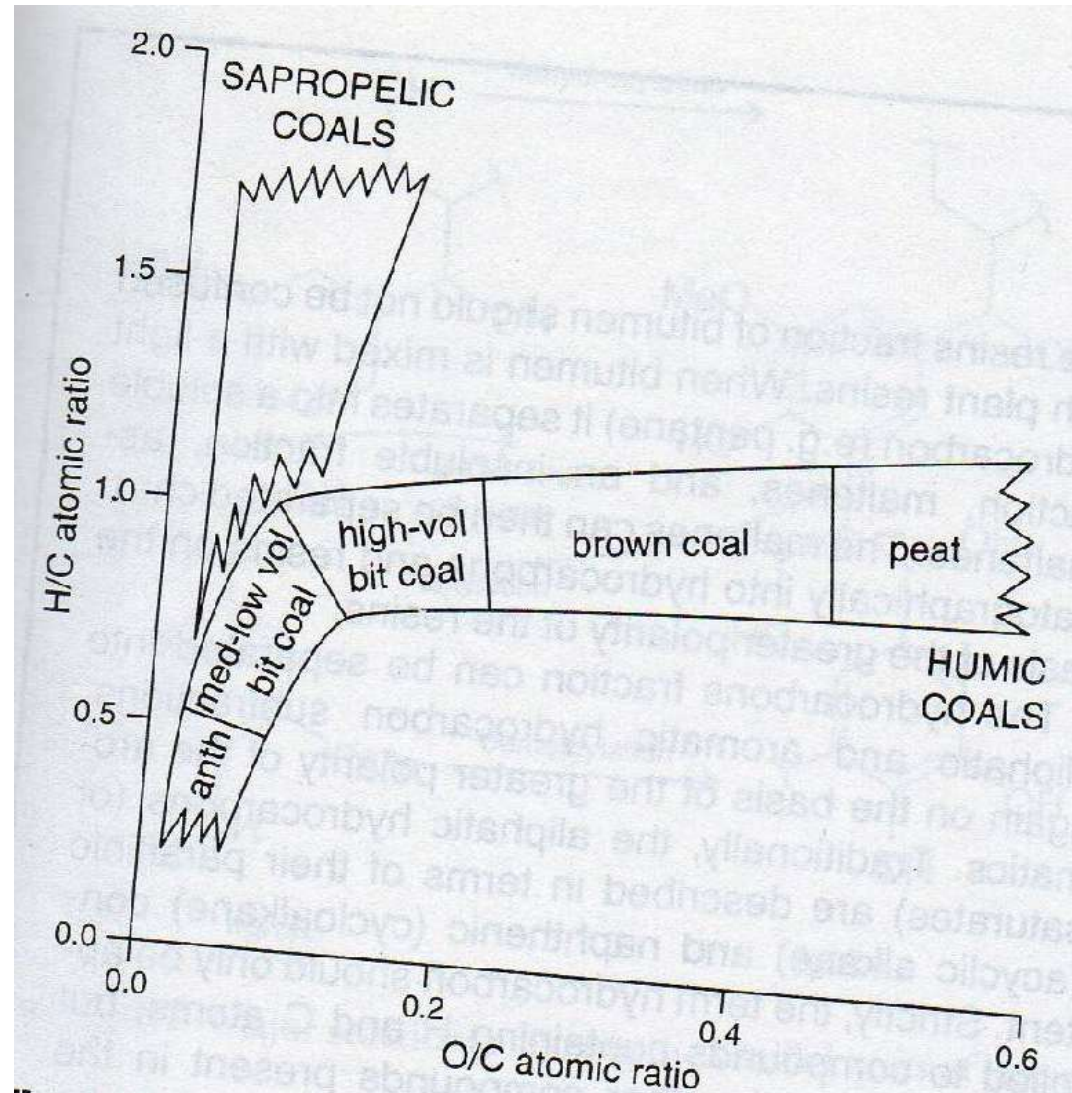
Classification of coal

Ranks of Coal	Fixed Carbon	Volatile Matter	Moisture
Peat	37	63	75
Lignite	29	26	46
Sub-bituminous	42	34	23
Low-rank/volatile bituminous	47	41	12
Medium-rank/volatile bituminous	54	41	5
High-rank/volatile bituminous	65	32	3
Low-rank/volatile semibituminous	75	22	3
Semi-anthracite	86	12	3
Anthracite	96	1.2	3

Schematic representation of formation of coal



Chemical composition of major coal macerals and plant tissues plotted on a van Krevelan diagram

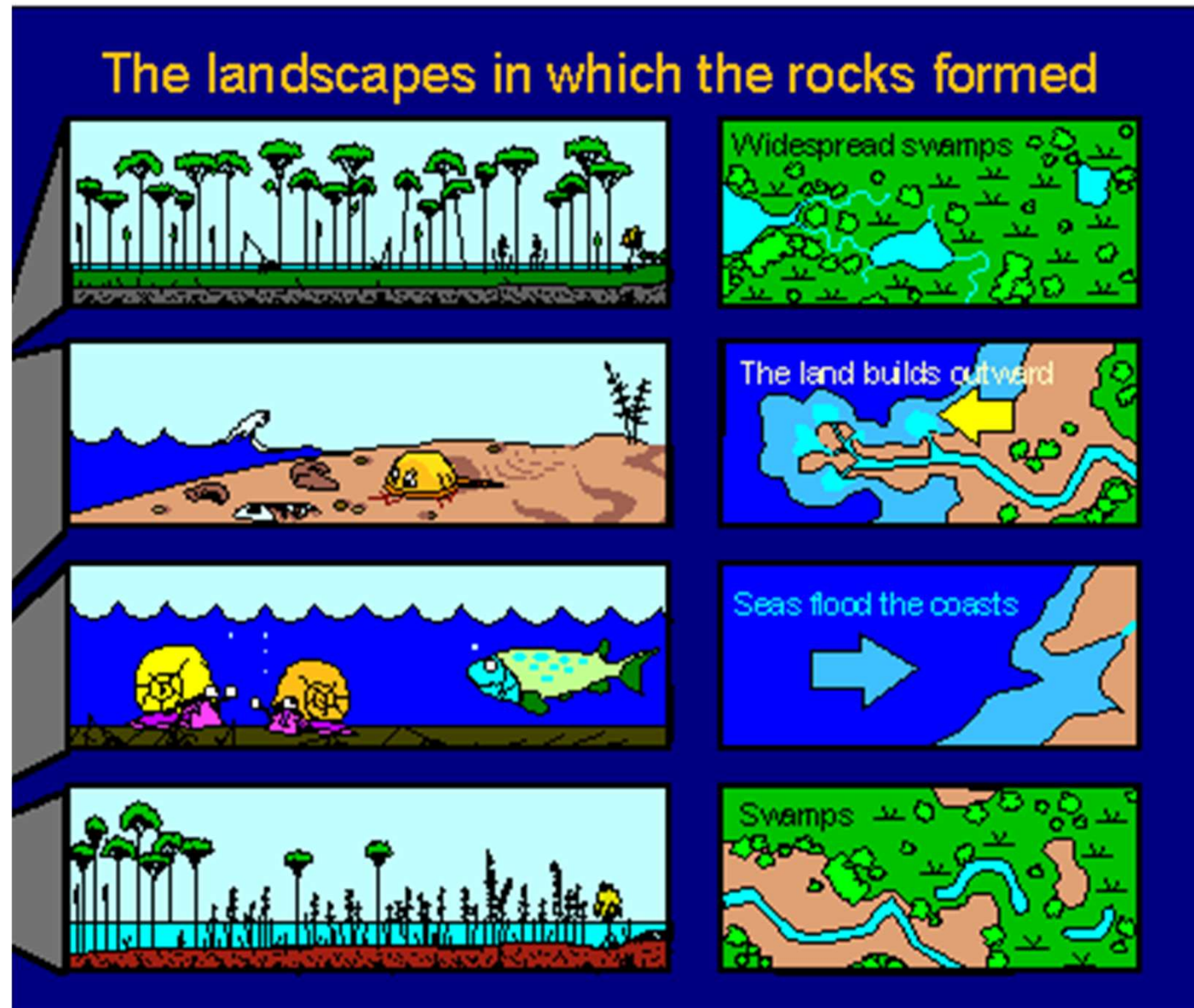


Temporal and Spatial distribution of Fossil organic carbon

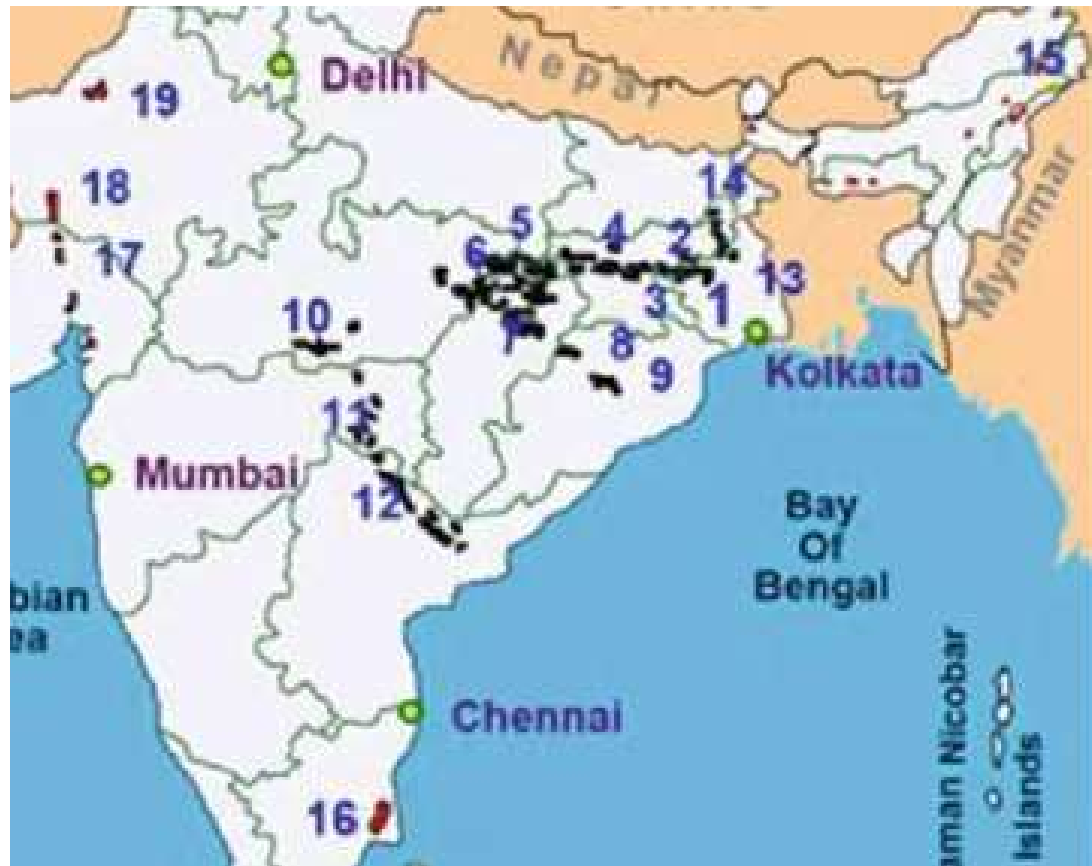
Coal and Kerogen

- **The episode of coal formation correlate with rising sea level**
- **Two major episodes of coal formation can be distinguished**
 - : Permian carboniferous**
 - : Jurassic to early tertiary**
- **The majority of early formed coal is bituminous/anthracite types, whereas tertiary coals are mainly brown coals**

Temporal and Spatial distribution of Fossil organic carbon



Coal deposits in India



GONDWANA COALFIELDS	
1. Raniganj	2. Jharia
3. Bokaro	4. North Karanpura
5. Singrauli	6. Sohagpur
7. Korba	8. Ib-valley
9. Talchir	10. Satpura
11. Wardha	12. Godavari
13. Birbhum	14. Rajmahal
TERTIARY COAL / LIGNITE FIELDS	
15. Assam-Mehgalaya	16. Neyveli
17. Cambay	18. Barmer-Sanchor
19. Bikaner	20. Jammu and Kashmir